

HAIML 2.0: A Human-Centered Framework for Cognitive Resilience and Human–AI Decision-Making

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Author Note

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Abstract

Artificial intelligence is increasingly affecting how people interpret information, generate alternatives, assess uncertainty, and make decisions. This expansion requires moving beyond tool proficiency and conventional AI literacy toward developing the human capabilities needed to maintain sound judgment in AI-mediated environments. HAIML 2.0 extends the Human-Centered AI Metacognitive Learning Model from its original educational foundation into a broader framework for cognitive resilience and human–AI decision-making. The model identifies four mutually reinforcing capacities: intentional AI engagement, metacognitive monitoring, calibrated trust, and ethical and accountable judgment. These capacities are proposed to strengthen cognitive resilience, defined as the sustained ability to reason independently, adapt under uncertainty, regulate reliance on AI, and preserve human agency while collaborating with increasingly capable systems. Education and professional preparation provide developmental environments in which individuals can practice and strengthen these capacities for application across increasingly complex or consequential decision contexts. HAIML 2.0 contributes a structured human–AI decision cycle, an expanded four-level engagement model, measurable constructs, and a research agenda for examining whether AI-assisted environments strengthen or erode human judgment. The framework is intended to support research, professional preparation, and the design of human-centered AI workflows across education, organizations, healthcare, public policy, cognitive security, and other decision-intensive domains.

Keywords: human-centered artificial intelligence; human–AI decision-making; cognitive resilience; metacognition; calibrated trust; human agency; cognitive security; AI in education

Version Relationship and Scholarly Positioning

HAIML 2.0 builds on the original Human-Centered AI Metacognitive Learning Model introduced by Reardon (2026). HAIML 1.0 established three interconnected areas: Experiential AI Use, Metacognitive Reflection, and Ethical Decision-Making, paired with four structured levels of AI engagement. HAIML 2.0 retains this educational foundation while extending its focus beyond classroom AI use to human–AI decision-making across educational, professional, organizational, and public information environments. This broader application introduces four connected capacities and positions cognitive resilience as the central developmental outcome.

The core contribution of HAIML 2.0 is the proposition that intentional AI engagement, metacognitive monitoring, calibrated trust, and ethical and accountable judgment work together to strengthen cognitive resilience. Together, these capacities are intended to help individuals maintain independent reasoning, adapt under uncertainty, regulate their reliance on AI, and remain responsible for decisions made in collaboration with increasingly capable AI systems.

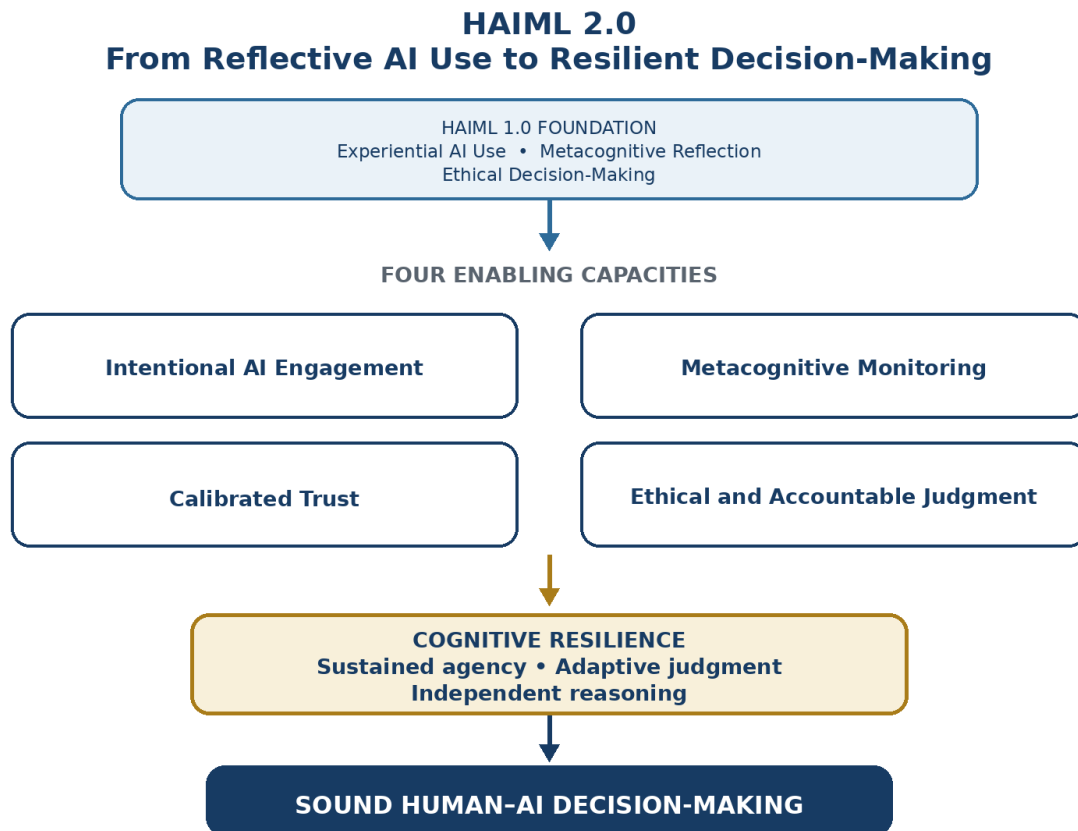


Figure 1. The HAIML 2.0 model. The original experiential, metacognitive, and ethical foundation develops four enabling capacities. Together, these capacities are proposed to strengthen cognitive resilience and support sound human–AI decision-making.

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The Emerging Human–AI Decision Environment

Generative and agentic AI systems do more than retrieve information. They can frame problems, suggest alternatives, summarize evidence, generate interpretations, and recommend actions. As a result, these systems may influence attention, confidence, memory, and judgment. The central question is no longer limited to whether an AI output is accurate. It must also consider whether the person remains involved in defining the problem, evaluating evidence, recognizing uncertainty, and accepting responsibility for the outcome.

This shift is especially important in higher education, where students learn how to reason under uncertainty. In addition to developing knowledge within their field, students learn to evaluate claims, consider tradeoffs, communicate decisions, and revise their judgments. When AI is used mainly as a productivity tool, it may reduce the thinking and effort needed to develop these capabilities. When AI is used intentionally, it can help students examine assumptions, test alternatives, receive feedback, and practice decision-making in ways that may be difficult to provide at scale. International guidance also emphasizes the importance of using generative AI in ways that preserve human agency and support the development of human capabilities in education (Miao & Holmes, 2023).

The same concerns apply in professional and public settings. An analyst may use AI to summarize threat information, a clinician may receive AI-assisted recommendations, a manager may use a system to rank options, and a member of the public may encounter AI-generated claims intended to influence beliefs or behavior. Across these settings, decision quality depends on more than the capabilities or accuracy of the AI system. It also depends on whether people can understand the context, recognize uncertainty, question recommendations, resist inappropriate influence, and remain responsible for the final decision. The challenge is to use AI in ways that support human judgment without allowing the system to gradually replace the thinking required for sound and accountable decisions.

From AI Literacy to Cognitive Resilience

AI literacy is necessary, but it is not enough. People should understand the capabilities, limitations, and risks of AI systems. However, knowing how a system works does not automatically lead to sound judgment when using it. A person may understand that an AI model can produce inaccurate or fabricated information and still accept a fluent response without checking it (Tankelevitch et al., 2024). Someone may also understand automation bias in theory but still defer to an AI recommendation when working under pressure (Parasuraman & Riley, 1997; Skitka et al., 1999). HAIML 2.0 therefore focuses on capacities that must be practiced, monitored, and strengthened over time.

Cognitive resilience is defined as the capacity to sustain independent reasoning, metacognitive regulation, adaptive judgment, calibrated trust in AI, and human agency while working with increasingly capable artificial intelligence systems. It does not require rejecting AI or completing every cognitive task independently. Instead, cognitive resilience includes recognizing when AI can appropriately reduce cognitive effort and when direct human thinking remains necessary. Cognitive offloading can be useful, but it may become harmful when it replaces the reasoning needed to develop or maintain knowledge and expertise (Risko & Gilbert, 2016). Cognitive resilience also involves knowing when and how to reenter the reasoning process if technology has begun to exert too much influence over how a problem is understood or a decision is made. It is therefore reflected in the ability to use AI assistance intentionally without surrendering the thinking, judgment, or responsibility required for the task. Within HAIML 2.0, cognitive resilience is conceptualized as a developmental capacity rather than a fixed individual trait. It can be strengthened through repeated opportunities to engage with AI intentionally, monitor one's own reasoning, evaluate AI outputs critically, calibrate reliance, and retain responsibility for final judgments.

Cognitive resilience has both protective and developmental values. The same capacities that support thoughtful everyday decision-making may also help people recognize manipulative framing, misinformation, persuasive technologies, and other influences that can shape or distort judgment (Waltzman, 2017). Within HAIML 2.0, cognitive security and resistance to manipulation are important applications of cognitive resilience, but they are not the primary purpose of the framework. The broader goal is to help people develop the capacity to maintain sound judgment across educational, professional, organizational, and public settings. Education and professional preparation provide structured environments in which people can practice questioning information, adjusting their reliance on AI, and returning to independent reasoning before they encounter decisions with more serious consequences. Over time, these capacities may transfer to less structured and higher-stakes situations in which people must recognize outside influence, evaluate uncertainty, and remain responsible for the choices they make.

How the Components of HAIML 2.0 Work Together

HAIML 2.0 brings together human-centered AI, judgment and decision-making, metacognition, cognitive resilience, and learning within a connected framework. Human-centered AI provides the foundation by emphasizing that AI systems and workflows should strengthen human capabilities, preserve meaningful human control, and support responsible engagement with AI (Shneiderman, 2020). Within this foundation, human–AI judgment and decision-making focuses on how people define problems, evaluate evidence, consider uncertainty, compare alternatives, and remain responsible for important choices. Metacognition supports this process by helping people recognize how AI may be influencing their confidence, attention, reasoning, and approach to making decisions (Flavell, 1979; Schraw & Dennison, 1994).

Cognitive resilience is the developmental outcome that HAIML 2.0 proposes can be strengthened as people learn to maintain or restore these capabilities through continued interactions with AI. Educational and professional settings provide structured opportunities to develop these capacities through practice, feedback, reflection, challenge, and the application of learning to new situations (Bjork et al., 2013; Zimmerman, 2002). This development does not occur through a single use of the decision cycle. Instead, people continue to monitor their thinking, evaluate whether they are relying on AI appropriately, and use what they learn to guide future interactions and decisions. By connecting these processes, HAIML 2.0 supports human–AI collaboration that strengthens human judgment and agency rather than allowing these capabilities to gradually weaken or be replaced.

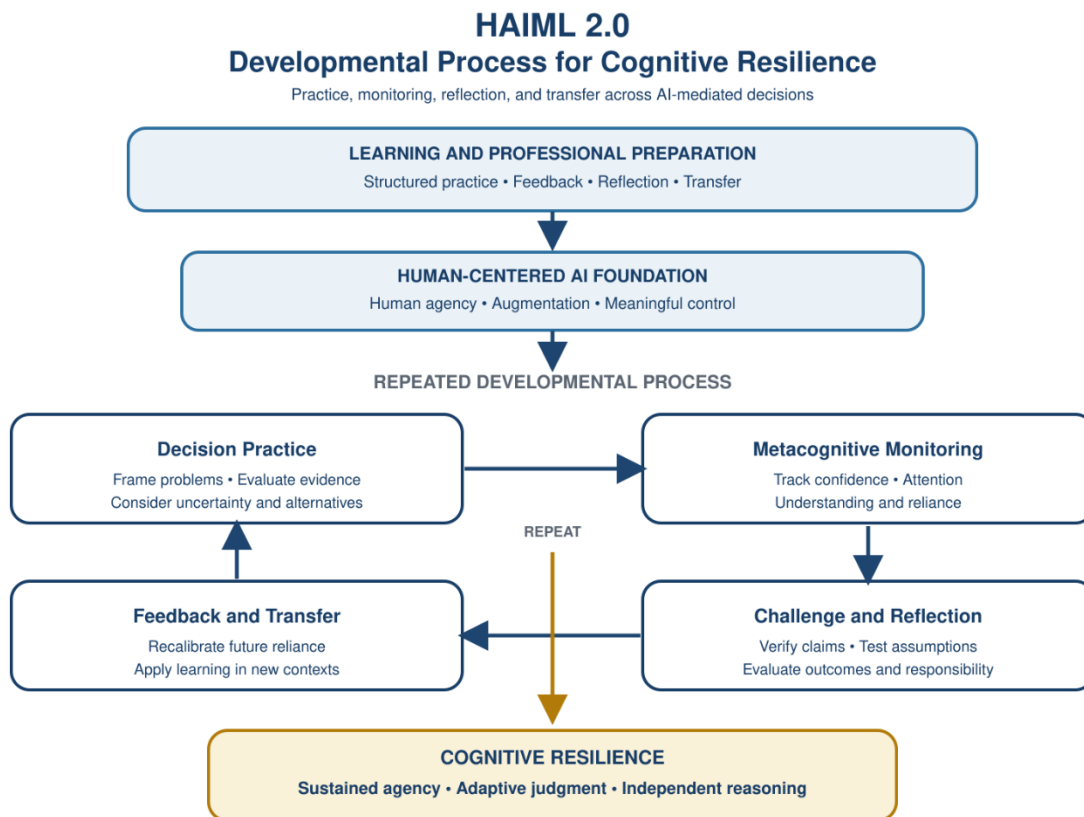


Figure 2. The HAIML 2.0 developmental process.

Learning and professional preparation provide structured opportunities to apply human-centered AI principles through repeated cycles of decision practice, metacognitive monitoring, challenge and reflection, and feedback and transfer. Through continued practice across AI-mediated decisions, these connected processes are proposed to strengthen cognitive resilience.

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Psychological and Human Factors Foundations

HAIML 2.0 draws from several established areas of psychology and human factors research. These foundations help explain how people regulate their behavior, evaluate information, manage cognitive effort, respond to automated recommendations, and make decisions under uncertainty. They also show why effective human–AI collaboration requires more than technical knowledge or access to capable systems. HAIML 2.0 does not propose new theories of agency, metacognition, trust, or decision-making. Instead, it brings these perspectives together to examine how human capabilities can be developed and maintained in AI-mediated environments.

Several research traditions are particularly relevant to this integration. Metacognitive research emphasizes the importance of monitoring and regulating one’s own thinking (Flavell, 1979; Nelson & Narens, 1990), while research on cognitive offloading demonstrates how people use external tools to reduce cognitive demands (Risko & Gilbert, 2016). Human factors research on automation has shown that people may misuse, disuse, or over-rely on automated systems and that appropriate reliance depends in part on how trust is calibrated to system capabilities and performance (Parasuraman & Riley, 1997; Lee & See, 2004). Research on judgment and decision-making further demonstrates that human reasoning is shaped by uncertainty, cognitive limitations, heuristics, and contextual influences. Together, these traditions provide a foundation for understanding why human–AI interaction requires not only knowledge of AI systems but also the capacity to monitor reasoning, regulate reliance, preserve human agency, and remain accountable for consequential decisions.

Human Agency and Self-Efficacy

Bandura’s social cognitive theory positions people as agents who act intentionally, anticipate consequences, regulate their behavior, and reflect on their functioning (Bandura, 2001). This perspective is foundational to HAIML because the framework rejects a passive model of AI use. People may receive suggestions, feedback, or decision support from AI, but they remain responsible for evaluating those contributions and determining how to act. Self-efficacy is also important because it influences whether individuals believe they can successfully exercise independent judgment (Bandura, 1997).

Although AI can scaffold performance and increase access, emerging research suggests that reliance on generative AI may alter cognitive effort and confidence during knowledge work (Lee et al., 2025). Over time, excessive dependence may limit opportunities for individuals to exercise independent reasoning and develop confidence in their own capabilities. HAIML therefore emphasizes AI use that supports human performance while preserving opportunities to develop confidence, competence, and decision ownership. From the perspective of HAIML 2.0, maintaining agency and confidence in one’s capacity to reason and decide are important conditions for cognitive resilience because individuals must be willing and able to reassert human judgment when AI assistance becomes unreliable, inappropriate, or overly influential.

Self-Regulated Learning and Metacognition

Self-regulated learners plan, monitor, and evaluate the strategies they use to accomplish a task (Zimmerman, 2002). Metacognition adds an awareness of one's knowledge and the ability to monitor and regulate cognitive processes (Schraw & Dennison, 1994). Flavell (1979) distinguished between possessing knowledge about one's cognition and actively monitoring that cognition while completing a task. Nelson and Narens (1990) further described metacognition in terms of monitoring and control. Individuals assess their own thinking and use that information to decide whether their approach needs to be adjusted. Together, these perspectives suggest that effective reasoning involves not only completing a task but also evaluating and adjusting the processes used to complete it.

AI introduces additional decisions about how people monitor and regulate their thinking. These include whether to consult the system, how to formulate a request, when to verify an output, how much weight to give a recommendation, and whether a fluent response reflects genuine understanding. These decisions can be difficult because generative AI may present accurate and inaccurate information using confident and coherent language, reducing the cues that would normally signal a need for verification (Tankelevitch et al., 2024). Research has also identified the possibility of “metacognitive laziness,” in which learners may reduce their own cognitive and metacognitive effort when working with generative AI (Fan et al., 2025). HAIML makes these decisions explicit and treats metacognitive monitoring as an ongoing part of effective human–AI interaction rather than an optional activity that occurs only after AI use. Within HAIML 2.0, this monitoring supports cognitive resilience by helping individuals recognize when their understanding, reasoning, or reliance on AI needs to be reassessed and adjusted.

Cognitive Offloading and Productive Effort

External tools can extend cognition, reduce working-memory demands, and allow people to direct their attention toward more complex activities. Cognition can be distributed across individuals, environments, and the tools they use (Hutchins, 1995), and some accounts propose that tools may become part of the cognitive process itself (Clark & Chalmers, 1998). Cognitive offloading is therefore not inherently harmful. It can support performance by shifting routine or lower-value cognitive demands to external resources. However, it may become problematic when it removes the mental processing needed to develop or maintain the capability being learned, practiced, or assessed (Risko & Gilbert, 2016; Sparrow et al., 2011). Research on desirable difficulties further demonstrates that making a task easier or completing it successfully does not necessarily produce durable learning (Bjork et al., 2013).

Generative AI increases both the potential value and the possible consequences of cognitive offloading because it can perform increasingly complex cognitive tasks rather than simply store or retrieve information. Emerging research suggests that frequent generative AI use may be associated with lower critical-thinking performance, with cognitive offloading offering one possible explanation for this relationship (Gerlich, 2025). A survey of knowledge workers

similarly found self-reported reductions in cognitive effort that varied according to whether users placed greater confidence in the AI system or in their own abilities (Lee et al., 2025). HAIML 2.0 therefore emphasizes productive cognitive effort, defined as the level and type of human thinking needed to develop, maintain, or use the capability required by a task. The framework evaluates AI assistance according to this cognitive purpose rather than assuming that either greater assistance or greater difficulty is inherently beneficial. From this perspective, cognitive resilience requires knowing not only when to use AI to reduce unnecessary effort but also when preserving human cognitive effort is necessary for learning, expertise, or sound judgment.

Dual-Process Theory, Automation Bias, and Calibrated Trust

Fluent and confident AI outputs may encourage rapid, intuitive acceptance, particularly when people are working under pressure, facing uncertainty, or lack relevant expertise. More deliberate evaluation requires individuals to slow down, identify assumptions, examine evidence, and consider alternative interpretations (Kahneman, 2011). Automation bias creates a related risk because people may overuse automated recommendations or fail to identify errors when a system appears reliable (Cummings, 2004; Parasuraman & Riley, 1997; Skitka et al., 1999). Bainbridge (1983) described this challenge as one of the “ironies of automation,” since increasingly capable automated systems may leave human overseers less prepared to intervene when intervention becomes necessary. These risks become particularly important in human–AI interaction because strong performance in one context may encourage reliance that extends beyond the conditions in which the system has demonstrated reliability.

The appropriate goal is therefore not maximum trust or minimum trust, but calibrated trust that reflects the system’s demonstrated performance within a specific task and context (Lee & See, 2004). Within HAIML 2.0, calibrated trust involves continuously evaluating whether the level of reliance placed on AI is warranted by the available evidence, the uncertainty of the situation, the consequences of the decision, and the potential costs of an incorrect output. Appropriate reliance may increase, decrease, or require additional verification as the task, context, or available evidence changes. HAIML treats trust calibration as an active and ongoing judgment rather than a stable attitude toward AI. This capacity contributes to cognitive resilience by helping individuals benefit from reliable AI assistance while remaining prepared to question, override, or disengage from the system when reliance is no longer justified.

Naturalistic Decision-Making and Situational Awareness

Real-world decisions often occur under time pressure, uncertainty, incomplete information, and changing conditions. Expertise in these environments involves recognizing meaningful patterns while remaining responsive to new or conflicting information (Klein, 1998). Situational awareness similarly requires individuals to perceive relevant elements in the environment, understand their meaning, and anticipate how a situation may develop over time (Endsley, 1995). Together, these perspectives emphasize that effective decision-making involves more than

selecting among available options. It also requires maintaining an accurate and evolving understanding of the situation as it develops.

AI can support these processes by identifying patterns, summarizing large amounts of information, highlighting relationships, and presenting possible outcomes. However, increased automation can also influence what information people attend to and how actively they remain involved in monitoring a situation (Bainbridge, 1983; Parasuraman et al., 2000). In human–AI decision-making, this creates a risk that the system’s representation of a problem may shape or narrow the human decision-maker’s understanding before alternative interpretations have been adequately considered. HAIML 2.0 therefore treats context, uncertainty, changing conditions, and potential consequences as essential features of human–AI decision-making. Maintaining cognitive resilience requires individuals to remain actively involved in framing the problem, updating their understanding as conditions change, and reconsidering AI-assisted interpretations when new or conflicting information emerges.

Cognitive Resilience as a Distinct Construct

HAIML 2.0 draws its central concept of resilience from established psychological literature, although it applies the term to a different set of conditions. Developmental and clinical psychology have traditionally examined resilience as the capacity to maintain or restore healthy functioning despite significant adversity. Masten (2001) described resilience as emerging from ordinary adaptive systems rather than from rare or extraordinary personal qualities. Bonanno (2004) similarly demonstrated that stable functioning following loss or trauma is more common than previously assumed. These accounts generally focus on adaptation during or following identifiable adverse experiences. Cognitive resilience within HAIML 2.0 instead refers to the sustained capacity to reason independently, regulate reliance on AI, adapt one’s judgment, and preserve human agency during ongoing collaboration with AI systems that are fluent and persuasive but not always reliable.

Cognitive resilience is related to, but distinct from, critical thinking, cognitive flexibility, and AI literacy. Critical thinking focuses on evaluating claims, evidence, assumptions, and conclusions, while cognitive flexibility concerns the ability to adapt to changing demands, perspectives, or conditions. AI literacy involves understanding AI capabilities, limitations, risks, and appropriate uses. Cognitive resilience incorporates elements of these competencies but focuses specifically on whether individuals can continue to use and coordinate them effectively during sustained AI-assisted decision-making. Within HAIML 2.0, cognitive resilience is therefore proposed as an integrative developmental outcome rather than as a replacement for any existing competence. Establishing it as a distinct construct will require empirical validation, including evidence that it can be differentiated from related constructs and demonstrated across varying levels of AI reliability, persuasiveness, uncertainty, and decision consequence.

Positioning HAIML 2.0 Relative to Adjacent Frameworks and Research Traditions

HAIML 2.0 draws from several established research traditions, making it important to clearly state its distinct contribution. Appropriate reliance, as described by Lee and See (2004), focuses on calibrating trust according to a system's demonstrated reliability. This calibration is necessary for effective human–AI collaboration, but it does not address the entire decision-making process. A person may place an appropriate level of trust in an AI system and still frame the problem poorly, overlook relevant evidence, fail to monitor changes in confidence, or avoid responsibility for the outcome. Calibrated trust is therefore one of HAIML 2.0's four enabling capacities rather than a complete account of cognitively resilient human–AI decision-making.

Levels-of-automation frameworks describe how tasks and functions may be divided between humans and machines (Parasuraman et al., 2000; Sheridan & Verplank, 1978). The HAIML engagement levels build on this tradition by identifying different degrees and forms of AI participation within a task or decision-making process. Many traditional automation frameworks were developed around clearly defined system functions. Generative and agentic AI, however, create a more continuous and conversational form of collaboration in which the roles of the human and AI may shift throughout an interaction. HAIML therefore pairs its engagement levels with four enabling capacities that address how individuals monitor, evaluate, and regulate their own participation as AI involvement changes. The engagement levels describe the degree and nature of AI participation, while the enabling capacities describe the quality of human judgment preserved throughout the process.

Metacognitive monitoring describes the human capacity to observe and regulate one's own cognitive processes (Flavell, 1979; Nelson & Narens, 1990; Schraw & Dennison, 1994). HAIML 2.0 does not propose a new theory of metacognition. Instead, it applies established metacognitive principles to the increasingly complex relationship between a person's own reasoning and an AI system's contribution to that reasoning. Users must assess not only what they know but also how AI may have influenced their confidence, attention, assumptions, and perceived understanding. Much of the foundational metacognitive literature predates sustained collaboration with systems capable of generating fluent explanations, interpretations, and recommendations. HAIML extends these principles into AI-mediated decision environments by making the monitoring of both human and AI contributions an explicit part of the decision process.

Human-centered AI provides another important basis for HAIML 2.0. Shneiderman (2020) describes human-centered AI as an approach to designing reliable and trustworthy systems that support rather than displace human control. Ethical approaches to AI also emphasize beneficence, avoiding harm, human autonomy, fairness, and transparency as important principles for responsible AI use (Floridi & Cowls, 2019). Together, these perspectives help explain how AI systems and workflows should be designed and used. HAIML 2.0 asks a related but different question about the capacities people need once AI becomes part of a learning, professional, or public decision environment. Even a well-designed system cannot ensure that a person will

frame a problem effectively, monitor reliance, challenge recommendations, or accept accountability for a consequential choice. Human-centered AI addresses the design of the system and interaction, while HAIML provides a developmental framework for the person participating in that interaction. The two perspectives are complementary because responsible human–AI collaboration depends on both supportive system design and capable human judgment.

HAIML 2.0 integrates these research traditions into a specific developmental plan. It proposes that intentional AI engagement, metacognitive monitoring, calibrated trust, and ethical and accountable judgment work together to strengthen cognitive resilience. No single capacity is expected to support cognitive resilience across the full range of human–AI decision-making demands. Their coordination is intended to help individuals remain engaged in problem framing, evidence evaluation, uncertainty recognition, decision ownership, and reflection across repeated interactions with AI. HAIML 2.0 therefore contributes an integrated framework for developing and evaluating the human capacities needed for sound human–AI decision-making.

Four Enabling Capacities and the Cognitive Resilience Outcome

HAIML 2.0 identifies four enabling capacities that support effective human participation in AI-mediated decision-making. These capacities describe what people need to do before, during, and after interacting with AI. Each capacity addresses a different risk associated with the growing role of AI in reasoning and decision processes. Although the capacities can be examined separately, HAIML proposes that their coordination is particularly important for cognitively resilient human–AI interaction. Together, they are expected to strengthen cognitive resilience by helping individuals maintain judgment, adapt their reliance, and preserve agency across changing tasks and contexts. Table 1 summarizes the four enabling capacities in HAIML 2.0 and describes the primary function of each capacity.

Table 1
Four Enabling Capacities in HAIML 2.0

Capacity	Observable behavior	Primary risk addressed
Intentional AI engagement	Decides whether and how to use AI based on the purpose of the task, what the person needs to learn, the potential consequences, and the expertise required.	Using AI without reflection and allowing it to replace thinking or the learning that the person should do.
Metacognitive monitoring	Pays attention to how AI use affects their confidence, understanding,	Believing they understand more than they do and becoming less aware of their own reasoning.

	attention, assumptions, and approach throughout the interaction.	
Calibrated trust	Decides how much to rely on AI by considering the available evidence, uncertainty, system limitations, previous performance, and opportunities to verify the output.	Automation bias, excessive trust, inappropriate reliance, and rejecting AI even when its use is warranted.
Ethical and accountable judgment	Considers stakeholders, fairness, transparency, authorship, potential consequences, decision ownership, and responsibility for the outcome.	Responsibility gaps, moral disengagement, unclear decision ownership, and harms that may go unrecognized.

The four capacities perform distinct but mutually reinforcing functions. Intentional AI engagement determines whether AI should be used and what role it should play within a particular task or decision. Metacognitive monitoring helps individuals recognize how the interaction may influence their confidence, understanding, attention, assumptions, and reasoning strategies. Calibrated trust helps individuals determine how much to rely on AI based on available evidence, uncertainty, system limitations, and the potential consequences of error. Ethical and accountable judgment broadens the evaluation beyond accuracy by considering stakeholders, potential harms, transparency, and responsibility for the outcome (Floridi & Cows, 2019). Cognitive resilience develops as individuals learn to coordinate these capacities across repeated AI interactions, changing conditions, and varying levels of uncertainty.

These capacities extend, rather than replace, the three-part structure introduced in HAIML 1.0 (Reardon, 2026). Experiential AI Use splits into intentional AI engagement and calibrated trust; Metacognitive Reflection becomes metacognitive monitoring; Ethical Decision-Making becomes ethical and accountable judgment. Cognitive resilience is the one genuinely new addition: HAIML 1.0 touched on resilience through its emphasis on agency, reflection, and ethical judgment, but did not treat it as a distinct, measurable outcome in its own right.

The HAIML Human–AI Decision Cycle

HAIML 2.0 operationalizes the framework through a repeatable Human–AI Decision Cycle. The cycle translates the four enabling capacities into actions that can be practiced, observed, and evaluated. It is designed to preserve meaningful human participation throughout an AI-assisted decision rather than limiting human oversight to a final review of the output. The cycle can be applied in academic assignments, simulations, professional workflows, decision-support

environments, and research protocols. Although the stages are presented sequentially for clarity, the cycle is iterative rather than strictly linear. Individuals may return to an earlier stage when new evidence, uncertainty, changing conditions, or ethical concerns emerge. Table 2 summarizes the seven stages of the Human–AI Decision Cycle and the human-centered action associated with each stage.

Table 2
HAIML 2.0 Human–AI Decision Cycle

Stage	Human-centered action
1. Frame	Define the decision, desired outcome, stakeholders, constraints, uncertainties, and level of consequence before consulting AI.
2. Position	Assign AI an explicit role within the task: human-only, AI-assisted, human–AI collaboration, or AI-integrated evaluation.
3. Engage	Use AI to generate information, alternatives, critique, analysis, or simulation while preserving an appropriate record of consequential inputs and outputs.
4. Monitor	Examine changes in confidence, assumptions, attention, perceived understanding, reasoning strategy, and willingness to defer to AI.
5. Challenge	Verify important claims, seek disconfirming evidence, compare alternatives, test uncertainty, and identify missing information or context.
6. Decide	Exercise human judgment, document the rationale for the decision, and identify the responsibilities that cannot be delegated to AI.
7. Reflect and Adapt	Evaluate the quality of both the outcome and the decision process, revise future reliance on AI, and transfer what was learned to new contexts.

The cycle is intended to prevent the evaluation process from being reduced to fact-checking at the end of an AI interaction. Instead, it distributes human participation across problem framing, role assignment, engagement, metacognitive monitoring, critical challenge, decision ownership,

and reflection. Each stage activates one or more of the four enabling capacities and is designed to keep the human decision-maker meaningfully involved throughout the process. This is important because an AI output can be technically accurate while still being based on a poorly framed problem, incomplete context, inappropriate assumptions, or insufficient consideration of stakeholders. A sound decision therefore depends on both the quality of the AI contribution and the quality of the human reasoning surrounding it. Through repeated use, the cycle is intended to help individuals develop more deliberate patterns of engagement, better calibrate their reliance on AI, and strengthen cognitive resilience across contexts.

Expanded Levels of Human–AI Engagement

HAIML 2.0 retains the four original AI Use Levels introduced in HAIML 1.0 while broadening their application beyond academic assignments and course policies (Reardon, 2026). The levels function as a companion structure to the four enabling capacities rather than as capacities themselves. The enabling capacities describe the quality of judgment a person brings to an AI-mediated task, while the engagement levels describe how cognitive work and decision authority are allocated between the person and the AI system. The two structures are intended to be used together, but neither can substitute for the other. A person may be permitted to use AI extensively and still exercise poor judgment, just as a person completing a task without AI may demonstrate careful reasoning and strong decision-making. The level of AI involvement therefore does not determine the quality of human judgment.

The expanded levels build on the levels-of-automation tradition introduced by Sheridan and Verplank (1978) and further developed by Parasuraman et al. (2000). Many traditional levels-of-automation models describe how relatively discrete functions are allocated between human operators and automated systems. Generative and agentic AI can create a more continuous and conversational distribution of cognitive work. The role of AI may shift throughout a task as the system generates alternatives, analyzes information, critiques reasoning, or supports evaluation. The HAIML engagement levels extend the earlier tradition by describing how people can intentionally allocate cognitive and decision responsibilities within these more fluid human–AI workflows. Table 3 summarizes the four HAIML 2.0 levels of engagement, the role assigned to AI at each level, the corresponding human responsibility, and the developmental purpose.

Table 3

HAIML 2.0 Levels of Engagement

Level	AI role	Human responsibility	Developmental purpose
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Level 1: Human-only	Makes no AI contribution to the focal task.	Performs and explains the reasoning independently.	Establishes baseline capability, foundational practice, and confidence in independent performance.
Level 2: AI-assisted	Clarifies, brainstorms, retrieves information, or provides limited feedback.	Defines the problem and completes the principal reasoning.	Supports exploration without surrendering the core cognitive work required for learning or expertise.
Level 3: Human–AI collaboration	Participates iteratively in drafting, analysis, critique, comparison, or alternative generation.	Evaluates, revises, rejects, and integrates AI contributions.	Develops judgment, comparison, revision, and trust calibration through active collaboration.
Level 4: AI-integrated evaluation	Supports complex workflows, simulations, recommendations, or multistage production.	Audits evidence, uncertainty, bias, consequences, system contributions, and final decisions.	Builds professional readiness and accountable AI use in complex or consequential contexts.

The levels should not be interpreted as a hierarchy in which greater AI involvement is automatically more advanced or desirable. The appropriate level depends on the purpose of the task, the person’s developmental stage and expertise, the consequences of error, the availability of verification, and the cognitive capability being developed or assessed. A foundational learning task may require Level 1 so that independent reasoning can be observed, while a complex professional simulation may appropriately require Level 4. The same individual may need to move between levels as the task, evidence, or level of risk changes. Cognitive resilience includes the ability to select an appropriate level of AI engagement and to adjust that level deliberately rather than defaulting to either maximum use or complete avoidance.

Challenges to Cognitive Resilience

HAIML 2.0 treats challenges to human judgment as design, learning, and organizational concerns rather than as evidence that people are inherently the weakest link in an AI system. Many failures emerge through interactions among human cognitive tendencies, characteristics of

AI systems, and the environments in which decisions are made. AI systems may amplify existing cognitive vulnerabilities by presenting information fluently, directing attention toward selected details, or appearing more certain than the available evidence warrants. These risks are also shaped by workload, time constraints, organizational expectations, system design, and access to meaningful verification.

Research on automation bias demonstrates that people may accept AI recommendations without adequately evaluating their accuracy (Parasuraman & Riley, 1997). Work on cognitive offloading shows that reliance on external systems can reduce the depth of reasoning individuals apply to a task (Risko & Gilbert, 2016). Studies of metacognitive demands in AI-assisted work suggest that generative systems can complicate people's ability to monitor their own understanding and calibrate their reliance appropriately (Tankelevitch et al., 2024). Research on information influence further shows that synthetic or targeted content can shape judgment through emotional appeal, repetition, and apparent authority (Waltzman, 2017). Together, this body of work indicates that these vulnerabilities can affect how people evaluate information and regulate their reliance on technological systems. Table 4 summarizes eight cognitive vulnerabilities that may arise during AI use, how each vulnerability may be expressed in AI-mediated contexts, and the corresponding HAIML response.

Table 4
HAIML 2.0 Cognitive Vulnerabilities and Responses

Vulnerability	AI-mediated expression	HAIML response
Automation bias	People may accept an AI recommendation without independently evaluating its accuracy, evidence, or appropriateness.	Adjust trust according to the system's demonstrated reliability and deliberately question important recommendations.
Cognitive offloading	People may delegate reasoning to AI that is necessary for developing knowledge, expertise, or ownership of a decision.	Determine the appropriate role of AI based on the purpose of the task and the cognitive skills that need to be developed or maintained.
Illusion of understanding	People may mistake a fluent AI explanation or immediate access to information for genuine understanding.	Ask individuals to explain the concept in their own words, apply it in a new context, and reflect on what they still do not understand.

Anchoring and framing	Early AI responses may shape how a problem is understood and limit the alternatives a person considers.	Define the problem before consulting AI and intentionally consider additional or competing alternatives.
Confidence distortion	AI may increase a person's confidence without improving accuracy or may reduce appropriate confidence in their own abilities.	Compare confidence with actual performance and periodically complete similar tasks without AI assistance.
Information manipulation	Synthetic or targeted content may influence judgment by appealing to emotion, identity, repetition, personalization, or apparent authority.	Pause to consider how the content is influencing judgment, evaluate the source, and verify the information within its broader context.
Responsibility diffusion	People may treat AI as responsible for a decision that still requires human judgment and ownership.	Clearly identify who owns the final decision and document the reasoning used to reach it.
Attentional narrowing	AI summaries, rankings, or alerts may direct attention toward certain information while causing other relevant details to be overlooked.	Maintain awareness of the broader situation and deliberately check for missing or excluded information.

These vulnerabilities are not independent of one another. Cognitive offloading may reduce the knowledge and reasoning needed to challenge an AI recommendation, while automation bias may increase the likelihood that the recommendation will be accepted. Anchoring may limit the alternatives considered, and confidence distortion may prevent a person from recognizing that the decision process has become constrained. Information manipulation can further exploit these conditions by presenting targeted content that appears credible, relevant, or emotionally compelling. In response, HAIML distributes human participation throughout the decision process so that monitoring, challenge, verification, and accountability do not depend on a single final review. The four enabling capacities are intended to help individuals recognize and respond to these interacting vulnerabilities, strengthening cognitive resilience across repeated and increasingly complex human–AI interactions.

Cognitive Resilience as an Educational and Professional Outcome

Within HAIML 2.0, cognitive resilience is conceptualized as a developmental and measurable outcome rather than as a general aspiration. It is proposed to develop through repeated cycles of intentional engagement, metacognitive monitoring, challenge, decision-making, reflection, and transfer. Learners and professionals should encounter AI that is helpful in some situations but uncertain, incomplete, biased, persuasive, or incorrect in others. Structured exposure to these varied conditions can provide opportunities to recognize changes in system reliability and adjust reliance accordingly. Emerging research suggests that AI dependence may be associated with lower critical-thinking performance through cognitive fatigue, with stronger AI literacy potentially reducing this relationship (Tian & Zhang, 2025). Related research indicates that critical-thinking disposition may predict more reflective and cautious reliance on generative AI during problem-solving (Hou et al., 2025). These findings do not establish the effectiveness of HAIML, but they support the need to examine how individual capabilities influence reliance and judgment in AI-mediated environments.

This approach has direct implications for how educational programs and professional fields prepare people to make decisions alongside AI. Understanding system capabilities and limitations remains important, but technical knowledge alone does not ensure sound judgment during use. Individuals must also learn to frame decisions, interpret uncertainty, evaluate evidence, recognize cognitive risks, allocate work between humans and AI, and preserve accountability. Training informed by HAIML 2.0 would evaluate both the performance of the AI system and the quality of the human reasoning surrounding its use. It would also examine whether these capacities transfer across tasks, AI systems, and levels of uncertainty rather than appearing only within the conditions in which they were initially practiced. The relevant outcome is therefore not simply whether AI improved speed or accuracy, but whether the combined human–AI process produced a decision that was sound, ethical, transparent, and defensible. Table 5 identifies the decision-maker capabilities emphasized in HAIML 2.0 and the observable behaviors associated with each capability.

Table 5

HAIML 2.0 Decision-Maker Capabilities

Decision-maker capability	Observed behavior
Decision framing	Clearly defines the goal, stakeholders, constraints, uncertainties, and possible consequences before selecting or consulting an AI tool.

Evidence evaluation	Distinguishes verified evidence from assumptions, interpretations, and AI-generated claims that have not been confirmed.
Trust calibration	Adjusts how much they rely on AI based on the system, the task, the level of uncertainty, and the system's demonstrated performance.
Bias and influence awareness	Recognizes cognitive biases, persuasive framing, missing information, and possible attempts to influence or manipulate judgment.
Human–AI workflow design	Determines which parts of a task should be completed by the person and which can be supported by AI based on their respective capabilities, limitations, the stakes involved, and the ability to verify the results.
Ethical accountability	Clearly identifies who owns the final decision and documents how transparency, stakeholder needs, possible harms, and the need for escalation were considered.
Cognitive resilience	Maintains or restores independent reasoning, metacognitive regulation, appropriate reliance on AI, adaptive judgment, and human agency when facing uncertainty, pressure, changing conditions, or information overload.

These indicators provide starting points for operationalizing cognitive resilience across learning and professional environments. No single behavior is sufficient to demonstrate the construct in its entirety. Evaluation should include both process and outcome measures because an accurate answer may result from inappropriate reliance, while a well-reasoned process may still produce an incorrect outcome under uncertainty. Performance should also be examined across changes in system reliability, task complexity, time pressure, consequence, and context to determine whether these capabilities transfer beyond the conditions in which they were initially developed. This approach would allow researchers to test whether HAIML-supported experiences strengthen durable human capabilities rather than simply improving performance while AI assistance is available.

HAIML as a Foundation for Decision-Making Across Fields

HAIML 2.0 is organized around human judgment and decision-making, rather than a single profession, technology, or institutional setting. Across fields, people use AI to interpret information, assess risks, compare options, and make decisions with different levels of consequence. Whether the setting involves healthcare, security, organizational leadership, public information, or education, the decision-maker must determine how AI should participate, evaluate its contributions, exercise judgment, and remain accountable for the outcome. HAIML 2.0 identifies these shared processes while recognizing that their application must be informed by the knowledge, standards, and responsibilities of each field. This structure allows the framework to support interdisciplinary research without treating all decision environments as equivalent.

Several interdisciplinary areas of knowledge and practice are particularly relevant to understanding and developing cognitively resilient human–AI decision-making:

- **Human-Centered AI Foundations:** Examines how AI can support human capabilities while preserving human agency, meaningful control, and clear boundaries between human and system responsibilities.
- **Psychology of Judgment and Decision-Making:** Explains how people make decisions under uncertainty and how heuristics, biases, confidence, expertise, and perceptions of risk influence judgment.
- **AI Systems for Decision Support:** Focuses on what AI systems can and cannot do, how their performance and outputs should be evaluated, and how they can be integrated responsibly into human workflows.
- **Metacognition and Cognitive Resilience:** Addresses how people monitor and regulate their thinking, determine when to rely on AI, adapt when conditions change, and apply what they have learned in new situations.
- **Cognitive Security:** Examines how misinformation, manipulation, persuasive technologies, synthetic content, and attention-shaping systems can influence human judgment and decision autonomy (Pitts, 2026; Waltzman, 2017). It also considers how source evaluation, contextual verification, metacognitive awareness, and cognitive resilience can help people recognize and respond to these influences.
- **Ethics, Governance, and Accountability:** Considers fairness, transparency, responsibility, oversight, escalation procedures, and the institutional safeguards needed to support responsible AI-assisted decisions.
- **Human Performance Under Pressure:** Examines how situational awareness, communication, fatigue, workload, and time pressure affect individual and team decision-making.

- **Applied Practice and Evaluation:** Uses simulations, red teaming, structured decision exercises, and realistic evaluation methods to examine how people make decisions with AI under changing or challenging conditions.

These areas are not sequential stages, nor will they receive the same emphasis in every field. Together, they establish the interdisciplinary foundation needed to understand how people engage with AI before, during, and after a decision. The four enabling capacities and the Human–AI Decision Cycle translate this foundation into a shared structure for research, practice, and evaluation. Within that structure, each field can define the evidence, expertise, safeguards, and standards appropriate to its decisions and consequences. HAIML 2.0 therefore provides a common vocabulary for studying cognitively resilient human–AI decision-making while preserving the contextual distinctions that make responsible application possible.

Measurement and Evaluation

Research examining HAIML 2.0 should evaluate both the quality of a decision and the process used to reach it. Accuracy alone cannot show whether a person understood the task, recognized uncertainty, relied too heavily on AI, or could explain the reasoning without AI assistance. A person may reach the correct decision simply by accepting an accurate AI recommendation without evaluating it. In contrast, a person may use careful reasoning and still reach an incorrect decision when the available evidence is limited or uncertain. Self-report measures are also not enough because a person’s confidence or perceived understanding may not match actual performance. Evaluation should therefore include behavioral observations, performance measures, decision-process evidence, and self-reported experiences. A multimethod approach is especially important for distinguishing constructs such as resistance to manipulation, which can be initially examined through controlled tasks, from constructs such as cognitive security, which require observation across repeated or extended exposure.

Table 6

HAIML 2.0 Measurement and Evaluation

Construct	What could be measured	How it could be evaluated
Independent reasoning	The quality, completeness, and independence of a person’s reasoning before and after consulting AI.	Use a defined rubric to compare the person’s initial reasoning with the final AI-supported response and determine whether the reasoning was strengthened, maintained, or weakened after AI consultation.

Construct	What could be measured	How it could be evaluated
Metacognitive monitoring and regulation	How accurately a person evaluates their confidence and understanding and whether they adjust their reasoning or reliance when they recognize uncertainty or gaps in understanding.	Compare confidence ratings with actual performance and examine whether the person changes their reasoning, verification, or reliance when confidence and accuracy do not match.
Appropriate reliance	Whether a person accepts accurate AI guidance and rejects inaccurate or inappropriate guidance.	Use controlled decision tasks in which the accuracy and appropriateness of AI recommendations vary, and examine whether the person accepts reliable guidance and rejects inaccurate or inappropriate guidance.
Adaptive judgment	Whether a person can reconsider and revise a judgment when evidence, constraints, AI reliability, or decision conditions change.	Introduce new, conflicting, or incomplete information during a decision scenario and examine whether the person appropriately maintains or revises the initial judgment.
Resistance to manipulation	Whether a person recognizes persuasive framing, missing information, synthetic evidence, emotional influence, or other manipulation tactics.	Use information scenarios containing misleading or manipulative elements and analyze how the person evaluates them.
Cognitive security	Whether a person can protect judgment, decision autonomy, and human agency when exposed to misinformation, synthetic content, persuasive technologies, or coordinated information influence.	Use realistic information environments containing credible and deceptive sources, synthetic content, and influence tactics. Examine how the person verifies information, recognizes attempts to shape judgment, and maintains or appropriately revises an independent decision.

Construct	What could be measured	How it could be evaluated
Ethical and accountable judgment	Whether a person identifies relevant stakeholders, possible harms, competing responsibilities, accountability concerns, and ownership of the final decision.	Use structured cases that require the person to identify ethical and accountability concerns and explain and justify the final decision.
Human agency and decision ownership	Whether a person maintains ownership of the final decision, can explain the reasoning behind it, and can distinguish their own contributions from those made by AI.	Review decision explanations and AI interaction records, followed by an interview or written reflection examining how the person understood and maintained their role in the decision.
Transfer	Whether a person can use the HAIML capacities and decision practices when facing a new task, setting, or type of problem.	Ask the person to apply what they learned first to a similar task and then to a different or unfamiliar decision situation.
Cognitive resilience	Whether a person can maintain or restore independent reasoning, metacognitive regulation, appropriate reliance, adaptive judgment, and human agency as AI reliability, uncertainty, pressure, or task complexity changes.	Observe the person across several AI-supported decisions in which system reliability, uncertainty, time pressure, available information, and task difficulty change. Examine whether independent reasoning, metacognitive regulation, appropriate reliance, adaptive judgment, and human agency are maintained or restored as conditions change.

These constructs should not be treated as different names for the same ability. Cognitive resilience is proposed as an integrative developmental outcome that depends on several related capacities working together. Research will need to determine whether cognitive resilience can be measured separately from critical thinking, AI literacy, cognitive flexibility, and general decision-making ability. Evaluation should also assess performance when AI assistance is reduced or unavailable to determine whether people retain and can use these capabilities without

immediate system support. Researchers should examine whether improvements continue over time and whether people can apply these capabilities beyond the original learning or training environment. This will require studies that follow participants across repeated tasks, changing conditions, and different decision contexts. Longitudinal and cross-domain research will be especially important for determining whether HAIML supports lasting human capabilities rather than short-term improvements that occur only while AI assistance is available.

Research Propositions

The following propositions turn the main ideas of HAIML 2.0 into relationships that can be tested through research. They can guide controlled studies, simulations, classroom research, and evaluations in professional settings. Each study will need to clearly define how HAIML is being used, what role AI plays, what type of decision participants are making, and what outcome is being measured. Researchers should also consider differences in AI reliability, task difficulty, participant expertise, time pressure, and the consequences of error. These propositions do not assume that HAIML will be effective in every setting. Instead, they provide starting points for testing, challenging, and refining the framework.

Table 7 presents twelve testable propositions derived from HAIML 2.0 and identifies the primary independent or predictor variables and dependent variables associated with each proposition. When a proposition includes several outcomes that measure the same underlying construct, such as the cognitive resilience areas in P4, the analysis will account for the increased possibility of finding a statistically significant result by chance across multiple tests. When the outcomes represent conceptually different dependent variables, such as rationale clarity and error-detection rate in P8, each outcome will be identified in advance and analyzed separately.

Table 7

HAIML 2.0 Testable Propositions

Proposition	Testable statement	Independent or predictor variable	Dependent variable and proposed measure
P1	Participants who use the structured HAIML Human–AI Decision Cycle will show a closer match between their confidence and actual accuracy than participants who use AI without structured monitoring and reflection.	AI-use condition: structured HAIML Human–AI Decision Cycle versus unstructured AI use without confidence or reflection prompts.	Confidence-accuracy calibration, measured by comparing participants' confidence ratings with their actual accuracy across trials.

Proposition	Testable statement	Independent or predictor variable	Dependent variable and proposed measure
P2	Participants who define the problem in writing before consulting AI will be less likely to anchor on the AI's initial framing and will generate more alternative solutions than participants who consult AI before defining the problem.	Timing of written problem definition: before the initial AI consultation versus after receiving the AI's initial response.	Anchoring, rated by trained evaluators using a defined rubric comparing the participant's final solution with the AI's initial framing, and the number of distinct alternative solutions generated. The two outcomes will be analyzed separately.
P3	Participants who practice with AI systems that vary in reliability and receive feedback on the outcomes will rely on AI more appropriately than participants who receive only a general warning about AI's limitations.	Training condition: practice with AI systems that vary in reliability and provide outcome feedback versus a general warning that AI can make mistakes and important outputs should be verified.	Appropriate reliance, measured by how often participants accept accurate AI recommendations and reject inaccurate recommendations.

Proposition	Testable statement	Independent or predictor variable	Dependent variable and proposed measure
P4	Participants who receive structured metacognitive prompts, including confidence ratings and self-explanation checkpoints, during HAIML practice will show greater improvement across the components of cognitive resilience than participants who complete the same practice without these prompts.	Metacognitive-prompt condition: confidence ratings and self-explanation checkpoints present versus absent.	Change from before to after the intervention across five areas of cognitive resilience: independent reasoning, metacognitive regulation, appropriate reliance, adaptive judgment, and maintenance of human agency. Each area will be measured separately using a clearly defined or validated measure. The analysis will account for the increased possibility of finding a statistically significant result by chance across multiple tests.
P5	Participants who are explicitly told that the human retains final decision ownership will be less likely to attribute responsibility for a poor outcome to the AI than participants who complete the same tasks without an explicit ownership statement.	Decision-ownership condition: explicit assignment of final decision ownership to the human present versus absent.	Participants' ratings of human and AI responsibility, each measured separately on a continuous scale from 0 to 100.
P6	Participants who practice across all four HAIML engagement levels will select a more appropriate role for AI on an unfamiliar transfer task than participants who spend the same amount of practice time at only one engagement level.	Range of engagement levels practiced: all four levels versus one specified level, with the number of practice trials and total practice time held constant.	Appropriateness of the AI role selected for an unfamiliar transfer task, evaluated by experts using a clearly defined and tested scoring rubric.

Proposition	Testable statement	Independent or predictor variable	Dependent variable and proposed measure
P7	After accounting for AI literacy and general critical-thinking ability, cognitive resilience scores will positively predict accuracy in identifying misleading or manipulative AI-generated content.	Cognitive resilience subscale scores, or an overall cognitive resilience score if research supports combining the subscales into one measure.	Accuracy in identifying misleading or manipulative AI-generated content after accounting for AI literacy and general critical-thinking ability.
P8	Teams using HAIML-structured workflows will produce decision rationales rated as clearer by independent reviewers and will detect more intentionally planted errors than teams using workflows focused on speed or output volume.	Workflow condition: HAIML-structured versus speed- or output-focused, with both conditions given equivalent tasks, information, and total time and required to provide a decision rationale.	Clarity of the decision rationale, rated by independent reviewers using a defined rubric, and the percentage of intentionally planted errors detected. The two outcomes will be analyzed separately.
P9	Participants who receive HAIML-based cognitive security practice across multiple sources and repeated exposure will maintain more accurate source evaluation and decision-making over time than participants who receive a single warning about misinformation or synthetic content.	Training condition: repeated HAIML-based cognitive security practice across multiple sources versus a single warning about misinformation or synthetic content.	Accuracy of source evaluation and decision-making at three or more points after training, including the extent to which performance is maintained or declines over time.

Proposition	Testable statement	Independent or predictor variable	Dependent variable and proposed measure
P10	Participants who use structured HAIML ethical and accountability prompts will identify more relevant stakeholders, potential harms, competing responsibilities, and accountability concerns than participants who evaluate the same AI-supported decisions without these prompts.	Prompt condition: structured ethical and accountability prompts present versus absent.	The number and quality of relevant stakeholders, potential harms, competing responsibilities, and accountability concerns identified, evaluated using a defined scoring process. The two outcomes will be analyzed separately.
P11	The effect of using the structured HAIML Human–AI Decision Cycle on decision accuracy will be smaller under high time pressure than under low time pressure.	AI-use condition: structured HAIML Human–AI Decision Cycle versus unstructured AI use, examined under constrained and unconstrained time conditions.	Decision accuracy under constrained and unconstrained time conditions, with the analysis examining whether time pressure changes the effect of using the structured Decision Cycle.
P12	HAIML-structured workflows will require more time per decision than comparable unstructured workflows.	Workflow condition: HAIML-structured versus unstructured, using the same decision tasks, information, and performance requirements.	Time needed to complete each decision, reported alongside accuracy and decision-quality outcomes to examine the time cost associated with structured HAIML use.

Together, these propositions examine confidence calibration, problem framing, appropriate reliance, metacognitive monitoring, responsibility, transfer, resistance to manipulation, team decision quality, cognitive security, ethical judgment, time pressure, and the time required for structured HAIML use. They can help researchers determine which parts of HAIML contribute most strongly to specific outcomes, as well as the conditions under which its benefits may be reduced or accompanied by additional costs.

Some propositions can be tested through controlled experiments, while others may require longitudinal studies, evaluations in real-world settings, or combinations of quantitative and qualitative methods. The results may show that HAIML works differently depending on a person's expertise, the population being studied, the decision context, or the type of AI system being used. Findings that do not support the proposed relationships will also be important because they can identify the framework's limitations and guide revisions to its constructs, relationships, and applications as HAIML 2.0 continues to develop.

Applications to Human–AI Decision-Making

The following domains show how HAIML 2.0 could be adapted for human–AI decision-making beyond the classroom. These applications provide starting points for research and practice rather than examples of settings in which the framework has already been validated. The four enabling capacities and the Human–AI Decision Cycle offer a common structure, but they will need to be adapted to the systems, evidence, expertise, regulations, and consequences within each field. Applications in regulated or high-consequence environments should be developed with domain experts, affected stakeholders, and appropriate oversight bodies. Research will be needed before conclusions can be drawn about whether HAIML is effective or can be applied successfully across these different settings.

In education and professional learning, HAIML can be used to structure when and how AI participates in learning, assessment, research, and skill development. This is important because generative AI introduces new metacognitive demands and, under some conditions, may reduce the cognitive and metacognitive effort learners devote to a task (Tankelevitch et al., 2024; Fan et al., 2025). Learners may first establish their own understanding or approach before consulting AI, compare their reasoning with AI-generated information, identify differences, and explain what they accepted, rejected, or revised. The engagement levels can also help instructors determine when independent performance is necessary and when AI-assisted or collaborative work better reflects the purpose of the task. Rather than evaluating AI use only according to whether it was permitted, HAIML shifts attention toward the quality of the learner's reasoning, monitoring, verification, and decision ownership. This approach may be especially useful for preparing learners to make decisions in professional environments where AI assistance will increasingly be available.

In professional and organizational decision-making, HAIML can provide a structure for determining what role AI should play within a workflow and where human judgment should remain explicit. Survey research with knowledge workers suggests that generative AI use may influence perceived cognitive effort and confidence, while human factors research has demonstrated the importance of calibrating reliance on automated systems (Lee et al., 2025; Lee & See, 2004). A manager using AI to compare resource-allocation options, for example, could define the decision and relevant constraints before consulting the system, evaluate the evidence supporting its recommendations, consider alternatives, and document the reasoning behind the

final choice. Similar practices could be adapted for analysts, researchers, communication professionals, and other knowledge workers who use AI to summarize information, generate alternatives, identify patterns, or support recommendations. In these settings, cognitive resilience involves more than detecting inaccurate AI outputs. It also requires recognizing when AI has begun to shape the framing of a problem, narrow the alternatives being considered, or influence confidence in ways that are not supported by the available evidence.

In high-consequence decision environments, calibrated trust, verification, situational awareness, and accountability become especially important. Research on automation has demonstrated the risks of inappropriate reliance, while research on situational awareness emphasizes the importance of maintaining an accurate and continually updated understanding of the environment in which decisions are made (Endsley, 1995; Parasuraman & Riley, 1997). AI-assisted recommendations in areas such as health care, public administration, security, or other regulated settings may involve decisions in which errors affect people beyond the immediate user. HAIML could provide a structure for examining whether decision-makers remain actively involved in framing the problem, evaluating uncertainty, challenging system recommendations, considering affected stakeholders, and identifying who retains responsibility for the final decision. Ethical and accountable judgment is particularly important in these settings because responsible AI-assisted decisions require attention to human autonomy, potential harms, fairness, transparency, and responsibility (Floridi & Cowls, 2019).

HAIML may also have applications in public information and cognitive security environments, where synthetic, personalized, persuasive, and AI-amplified information can shape how people interpret events and make decisions. These conditions create concerns for cognitive security and the protection of human judgment (Pitts, 2026; Waltzman, 2017). Pitts (2026) identifies decision autonomy as central to cognitive warfare and the growing role of cognitive security. Protecting decision autonomy requires preserving a person's ability to evaluate information, recognize attempts to influence judgment, and remain actively involved in deciding what to believe or do. In these contexts, cognitive resilience includes examining sources and framing, recognizing possible manipulation, identifying missing information, seeking additional evidence, and reconsidering conclusions when new information becomes available. The goal is not to make people resistant to all persuasion or distrustful of AI-generated information. Instead, HAIML emphasizes maintaining enough awareness and independent judgment to recognize when information may be shaping reasoning in ways that require further evaluation.

Across these applications, HAIML 2.0 does not prescribe identical procedures for every field. Instead, it provides a common approach for asking who is doing the cognitive work, how reliance on AI is being regulated, whether important evidence and alternatives are being considered, and who remains responsible for the outcome. The specific evidence, professional standards, safeguards, and acceptable levels of AI involvement must be determined within each domain. The four enabling capacities and the Human–AI Decision Cycle can be adapted to these differences without assuming that the same practices or measures will work equally well

everywhere. HAIML 2.0 therefore offers a shared framework for human–AI decision-making while recognizing that responsible application depends on the expertise, risks, requirements, and consequences of each setting.

Organizational Decision Support

Organizations increasingly use AI to support hiring, resource allocation, risk assessment, strategic planning, performance evaluation, and other decisions that affect individuals and groups. HAIML can help organizations clarify who is responsible for a decision, which AI contributions need to be verified, and how the reasoning behind the final decision should be documented. Intentional AI engagement can help decision-makers determine whether AI should generate alternatives, analyze information, rank options, or recommend an action. Metacognitive monitoring and calibrated trust can help people recognize when confidently presented outputs, efficiency, or the apparent objectivity of an AI system influence how much they rely on it. Appropriate reliance requires people to adjust their trust according to the capabilities and demonstrated performance of the system rather than assuming that greater automation should lead to greater trust (Lee & See, 2004). Ethical and accountable judgment also requires consideration of fairness, transparency, stakeholder impact, potential harm, and responsibility when AI contributes to an important organizational decision (Floridi & Cowsls, 2019).

HAIML can also guide the design and evaluation of AI-assisted organizational workflows. Decision-makers can define the purpose and consequences of a decision before selecting an AI role, identify which parts of the process require human judgment, and establish points at which AI-generated information should be challenged, verified, or escalated for additional review. Organizations can examine whether time pressure, workload, performance expectations, or established procedures encourage employees to rely on AI more heavily than the evidence warrants. They should also consider whether employees have the authority and practical ability to question an AI-supported recommendation without facing pressure to accept it. Documentation of consequential decisions can help distinguish AI contributions from human reasoning and make responsibility for the final outcome more visible. In this setting, HAIML supports evaluation of both the quality of the final decision and the human–AI process used to reach it, while recognizing that standards for oversight, documentation, escalation, and acceptable AI involvement will depend on the organization and the consequences of the decision.

Healthcare and Other High-Consequence Domains

In healthcare and other high-consequence settings, AI-assisted decisions may be made under time pressure, uncertainty, incomplete information, and significant risk. Situational awareness is especially important when conditions change quickly and decision-makers must continually update their understanding of what is happening (Endsley, 1995). HAIML emphasizes the need to consider the relevant context, verify important information, apply professional expertise, clearly define responsibility, and recognize when a decision requires escalation. The framework does not replace clinical judgment, professional standards, regulations, validated safety

procedures, or established requirements for human oversight. Instead, it provides a structure for examining how people use decision-support systems and how workload, fatigue, urgency, or a confidently presented AI recommendation may influence their reliance. Research on automation has shown that people may misuse or rely too heavily on automated systems, making calibrated trust especially important when errors can have serious consequences (Parasuraman & Riley, 1997; Lee & See, 2004).

The Human–AI Decision Cycle can help decision-makers determine when a recommendation should be questioned, when additional evidence is needed, and when further human review or escalation is necessary. In healthcare, this may involve examining whether relevant patient information, uncertainty, competing interpretations, and the limitations of an AI-assisted recommendation have been adequately considered before a decision is made. Similar principles may apply in defense, public safety, security, emergency management, infrastructure, and other settings where AI-assisted decisions can affect individuals, organizations, or communities. These environments may involve rapidly changing conditions, significant uncertainty, and serious consequences, making meaningful human control especially important. Standards for verification, escalation, documentation, and human authority must be established by experts within each field. HAIML 2.0 provides a structure for examining human judgment in these environments without prescribing how specialized or domain-specific decisions should be made.

Public Information Environments

HAIML 2.0 may also help people make more informed decisions about the information they encounter online. Digital platforms use behavioral data to shape what people see, which can influence what receives their attention, what they trust, and what they may eventually believe (Zuboff, 2019). Information environments can also be intentionally shaped to influence how people understand events and make decisions, creating broader concerns for cognitive security and the protection of human judgment (Waltzman, 2017). Generative AI adds to this challenge by making realistic text, images, audio, and video easier to create and distribute. The concern is not limited to whether information is true or false. The ways information is selected, framed, repeated, personalized, and presented can influence what people notice, how confident they feel, and which alternatives they consider.

In these environments, cognitive resilience involves recognizing when information may be influencing judgment and knowing when additional evaluation is needed. People may need to pause, examine the source, consider the original and broader context, identify missing information, seek additional evidence, and consider alternative explanations before accepting or sharing a claim. Metacognitive monitoring can help people recognize changes in their confidence, attention, or emotional reactions. Calibrated trust can help them determine how much weight to give a source, platform, or piece of AI-generated content based on available evidence rather than familiarity, repetition, or persuasive presentation. These practices may support cognitive security by helping people recognize and respond to misleading or

manipulative information while preserving decision autonomy. HAIML is not intended to tell people what to believe. Instead, it helps them examine how information may be influencing their reasoning, how they evaluate evidence, and how they reach conclusions.

Cognitive Security as a Specialized Application

Cognitive security is a specialized application of HAIML 2.0 rather than the primary purpose of the framework. Waltzman (2017) identified the weaponization of information as a growing security concern because digital platforms and other technologies can be used to influence beliefs, behavior, and decisions on a large scale. This perspective highlights why traditional cybersecurity protections may not be sufficient when manipulation is directed at human attention, reasoning, and decision-making. Pitts (2026) positions the protection of decision autonomy as central to cognitive security. From this perspective, the concern is not limited to whether people encounter false or manipulative information. It also involves whether they remain able to understand a situation, consider alternatives, evaluate evidence, and make decisions without covert or inappropriate influence.

Within HAIML 2.0, cognitive resilience is the human capability the framework seeks to develop, while cognitive security is one area in which that capability may help preserve decision autonomy. HAIML supports this capability by helping people recognize how AI and other information systems may influence their attention, confidence, reasoning, and willingness to accept system-generated recommendations without sufficient independent evaluation. The framework identifies capacities and decision practices that may help people recognize, evaluate, and respond to deceptive information, synthetic content, emotional manipulation, and other efforts to shape decisions within AI-mediated information environments. These practices do not prevent all forms of influence, nor do they assume that people can identify every attempt at manipulation. Instead, they place the preservation of human agency and judgment at the center of cognitively secure human–AI interaction.

Cognitive security concerns may involve synthetic media, persuasive AI, targeted messaging, information operations, and deliberate efforts to shape beliefs or behavior. In these situations, the Human–AI Decision Cycle can help people pause, reconsider how a problem or claim has been presented, examine the source and its context, assess uncertainty, seek conflicting evidence, and explain the basis for their decision. The framework may also help people distinguish between AI assistance that supports their reasoning and AI use that narrows their attention or encourages uncritical acceptance. However, individual cognitive resilience cannot replace responsible system design, institutional safeguards, methods for verifying the origin and authenticity of media, or coordinated responses to organized manipulation. Research will be needed to determine whether practicing HAIML helps people maintain control over their decisions and recognize manipulation without becoming overly distrustful, cognitively inflexible, or mentally overwhelmed.

Design Principles for Human-Centered AI Systems

HAIML 2.0 can inform how AI systems and human–AI workflows are designed, not only how people are taught to use them. Individuals should not be expected to maintain sound judgment while using systems that hide uncertainty, encourage immediate acceptance, or make responsibility unclear. Human-centered AI emphasizes the importance of designing systems that support human control while remaining reliable and trustworthy (Shneiderman, 2020). System design can either support or weaken intentional AI engagement, metacognitive monitoring, calibrated trust, and ethical and accountable judgment. Responsible AI design should also consider human autonomy, potential harms, fairness, and transparency when determining how people interact with and make decisions using AI (Floridi & Cowls, 2019). Design choices can also affect when and how much people rely on automated recommendations (Lee & See, 2004). AI systems and workflows should therefore help people remain involved in defining the problem, evaluating evidence, questioning recommendations, and making or approving the final decision. The following principles identify design features that may support cognitive resilience during human–AI interaction:

- **Allow people to define the problem first.** When a decision involves uncertainty, serious consequences, or competing stakeholder needs, the system should not determine how the problem is framed before the person has considered the purpose, context, and relevant constraints.
- **Clearly communicate uncertainty.** Systems should communicate relevant uncertainty, known limitations, and the evidence supporting an output in ways that people can understand and use.
- **Make it easier to consider other possibilities.** Systems should support the comparison of alternatives, explanations, counterarguments, and evidence that may challenge the initial recommendation.
- **Make AI contributions and sources visible.** People should be able to identify which information, recommendations, or interpretations came from AI, what sources were provided or used when that information is available, and how those contributions influenced the final decision.
- **Support source and content verification.** When systems generate, summarize, or modify information, they should help people examine its origin, context, and authenticity. AI-generated or altered content should be distinguishable from original source material when that distinction is relevant to the decision.
- **Create a deliberate pause when needed.** Systems should slow the process when automatic continuation, persuasive presentation, or immediate acceptance could create significant risk. These pauses should support reflection without creating unnecessary delay or cognitive burden.

- **Preserve decision autonomy.** Interface features, defaults, and recommendations should not pressure people to accept an output or make it unnecessarily difficult to consider alternatives. People should retain meaningful control over consequential judgments and actions.
- **Support additional review, escalation, and human override.** People should be able to question, reject, or escalate an AI recommendation when the evidence is limited, the consequences are serious, or the system's performance is uncertain.
- **Keep responsibility clear.** The workflow should identify which judgments and responsibilities remain with the person making or approving the final decision and when additional oversight is required.
- **Evaluate people, AI, and workflows together.** AI performance should be examined within the actual workflow, including the people using the system, the decisions being made, the organizational pressures affecting its use, and the environment in which it operates.
- **Examine effects across people and groups.** Evaluation should consider whether the system's recommendations, errors, explanations, or opportunities for human review affect users and stakeholders differently.
- **Examine effects over time.** Evaluation should consider whether repeated AI use strengthens or weakens human skill, confidence, independent performance, appropriate reliance, and the ability to apply judgment in new situations.

These principles are starting points for system and workflow design rather than requirements that have already been validated. Their value will depend on the task, the people using the system, their level of expertise, the consequences of error, and whether important information can be verified. A feature that encourages careful reflection in one setting may create unnecessary delay or cognitive burden in another. Evaluation should therefore consider the immediate quality of the decision, the fairness and accountability of the process, and the longer-term effects of AI use on human capabilities. A human-centered system informed by HAIML should ultimately be evaluated according to whether it helps people remain capable, adaptable, and accountable while preserving meaningful control over decisions made with AI.

Boundaries and Limitations

HAIML 2.0 is a conceptual and developmental framework. It does not claim that reflection, monitoring, or structured AI use can eliminate cognitive bias or guarantee sound decisions. Cognitive resilience may be influenced by individual differences, prior knowledge, expertise, workload, organizational culture, incentives, system design, and broader social conditions. The framework will need to be tested across different populations, decision settings, fields, and types of AI systems. Researchers will also need to define and measure its constructs carefully so that

cognitive resilience can be distinguished from related constructs such as critical thinking, cognitive flexibility, self-regulation, and AI literacy. Until sufficient research has been conducted, claims about the effectiveness of HAIML, its transfer across settings, and its long-term effects should remain cautious.

Cognitive resilience should not be treated only as an individual responsibility when institutions or technologies create predictable risks to human judgment. People cannot be expected to compensate for systems that hide uncertainty, workflows that reward speed without careful evaluation, or organizational cultures that discourage questioning and escalation. Human-centered AI requires attention to how systems and workflows support human control and responsibility rather than focusing only on the capabilities of the technology itself (Shneiderman, 2020). Responsibility must be shared among decision-makers, system designers and developers, organizations that deploy AI, policymakers, and the leaders who establish the conditions under which it is used. Future research should examine how these factors work together and whether specific environments support or weaken the capacities HAIML seeks to develop. This broader view helps prevent cognitive resilience from becoming a way to blame individuals for problems created by system design, organizational incentives, or institutional practices.

HAIML 2.0 also does not assume that human judgment is always better than AI output. People can make biased, inconsistent, or incorrect judgments, and AI systems can produce outputs with the same problems. The central question is how their different capabilities can be combined and evaluated so that errors are easier to identify, responsibility remains clear, and final decisions can be explained and defended. HAIML cannot replace technical security controls, professional expertise, institutional safeguards, legal requirements, or domain-specific standards for human oversight. In education, one implication is that AI detection should not be treated as the primary response to questions of authorship or academic integrity because detection-based approaches remain limited as AI-assisted work becomes more common (Eaton, 2023). A more sustainable approach is to build judgment, transparency, verification, and clear ownership into the process rather than relying mainly on attempts to detect inappropriate AI use after it occurs.

Conclusion

The future of AI should not be defined only by what increasingly capable systems can do. It should also be considered in terms of the human abilities that are strengthened, preserved, or weakened through their use. As AI becomes more involved in interpreting information, generating alternatives, assessing uncertainty, and recommending actions, knowing how to use the technology will not be enough. People will also need to recognize how AI is shaping their thinking, adjust how much they trust it, question incomplete or persuasive responses, and remain responsible for important decisions. These capacities matter not because human judgment is always better than AI, but because people still need to understand context, consider consequences, and take responsibility for the final choice.

HAIML 2.0 brings these needs together within a developmental framework for cognitive resilience and human–AI decision-making. It identifies intentional AI engagement, metacognitive monitoring, calibrated trust, and ethical and accountable judgment as four capacities that work together to support cognitive resilience. The Human–AI Decision Cycle turns these capacities into a process that can be practiced, observed, evaluated, and improved over time. Higher education provides an important setting for developing these capacities, but they are also relevant to professional practice, organizational decision support, healthcare, public information environments, cognitive security, and other settings in which AI may influence human judgment. HAIML also provides a research agenda for examining whether AI-assisted environments strengthen human capabilities or contribute to dependence, overconfidence, reduced reflection, and unclear responsibility. Empirical research will be needed to test these relationships and refine the framework across different populations, AI systems, and decision contexts.

AI may contribute to decisions by generating alternatives, identifying patterns, summarizing evidence, and recommending possible actions. However, AI systems cannot assume responsibility for how those contributions are interpreted or used. Sound human–AI decision-making requires people to remain involved in defining the problem, evaluating evidence, recognizing uncertainty, considering who may be affected, and explaining the final decision. HAIML 2.0 provides a structure for developing and studying these capacities while recognizing that the systems, safeguards, and standards required will differ across settings. Its central argument is that as AI capabilities expand, equal attention must be given to strengthening human agency, judgment, accountability, and cognitive resilience.

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